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Towards a Better Understanding of the Influence of Carbohydrates on the Structural, Dynamic, and Functional Properties of Fibromodulin: Crucial Insights From a Comparative In Silico Approach

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ABSTRACT

Proteoglycans are a class of biological macromolecules that are characterized by the presence of long chains of negatively charged glycosaminoglycans (GAGs). One family of proteoglycans, the small leucine-rich proteoglycans (SLRPs), is a class of well-conserved proteins, which are predominantly present in the extracellular matrix and are critical components of numerous processes in the extracellular matrix. Interestingly, fibromodulin (FMOD), one of the members of the SLRP family exists both as a proteoglycan and as a glycoprotein in juveniles and old-aged individuals, respectively. Despite their physiological significance and interesting variations in their carbohydrate composition, their structural dynamics (e.g., evolution of their structure over time), and their 3D conformational preferences remains unexplored to date. In this study, molecular modeling techniques were used to investigate the effects of changes in carbohydrate composition on fibromodulin structure and dynamics. A total of 6μs of all-atom molecular dynamics simulations were performed on three types of fibromodulin systems: non-glycosylated FMOD, the glycoprotein form lacking charged keratan sulfate disaccharides and fibromodulin proteoglycan with keratan sulfate GAGs. Substantial changes were found in FMOD dynamics between the three states. These changes were accompanied by modifications to the curved-solenoid architecture of FMOD, as well as local changes in solvent accessibilities to some critical residues. A relatively rigid, negatively charged motif, with a well-conserved sequence across various FMOD species, was identified in the unstructured N-terminal domain. This motif was highly accessible in both the proteoglycan and glycoprotein states, despite its proximity to carbohydrate chains. These two features (rigididy and high accessibility) suggest a key role for this motif in the protein-protein interactions established by Fibromodulin. Overall, the study provides insight into how carbohydrate composition influences protein dynamics and functions.

1 | Introduction

The extracellular matrix (ECM) constitutes a higly dynamic three-dimensional network that plays key roles in both the structural integrity of cells as well as in biological pathologies. This ECM is constituted of numerous macromolecules among them glycosaminoglycans and proteoglycans [1]. This latter class of molecules is implicated for instance in numerous actions in cancer biology [2] and their forms and functions are assessed in development, angiogenesis, cancer, autophagy,

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genetic diseases to name a few [3, 4]. Among proteoglycans, molecules classified as Small Leucine-Rich Proteoglycan (SLRP) are divided into different classes [5]: in class II, Fibromodulin (FMOD) proteoglycan remains uninvestigated from dynamical point of view. The members of this family are typically extracellular matrix proteoglycans, many of which have the ability to bind to collagen triple helices. FMOD is demonstrated to interact with collagens and play a role in the formation of large-diameter collagen fibrils during fibrillogenesis [6]. In addition to fibrillogenesis, FMOD is also observed to be important for maintenance of normal diameter and organization of collagen fibrils [7]. It performs its role in fibrillogenesis through interactions with the collagen cross-linking enzyme Lysyl Oxidase (LOX) [8]. FMOD is also involved in processes such as cellular differentiation, wound healing and angiogenesis [9–11]. It is also implicated in cancers, where it is suggested to have protumorigenic effects by promoting angiogenesis [12], or in disorders such as arthritis, where FMOD is found to be degraded proteolytically by enzymes such as MMP-13 [13–15].

The experimental 3D structure of FMOD was previously solved by X-ray crystallography [16]. FMOD was found to adopt the prototypical SLRP fold. This fold consists of a curved-solenoid architecture formed by multiple leucine-rich repeat (LRR) domains. The LRR domains are flanked by conserved cysteine disulfide bridges on both sides. The disordered N-terminal region consists of seven modified tyrosine sulfate residues. These residues are suggested to be involved in various protein-protein interactions via binding to motifs containing positively charged amino acids [17]. Though FMOD shares high degree of sequence identity with many other SLRP members, it does not form dimers like other SLRP members, such as Decorin and Osteoadherin [16]. FMOD comprises 4 N-glycosylation sites, which are substituted with keratan sulfate-1 Glycosaminoglycans (GAGs) [13].

Interestingly, experimental studies have found that the carbohydrate composition of FMOD shows an age-dependent variation. During development and in juvenile individuals, FMOD was found as a proteoglycan, with long keratan sulfate chains in vertebral tissues as well as in cartilage. In aged individuals however, FMOD lacked glycosaminoglycans and was instead found as a glycoprotein, substituted with N-glycan chains [13, 18]. Furthermore, in arthritic tissues, FMOD was observed in a degraded form, possibly due to the proteolytic action by enzymes such as MMP-13 [14, 15, 18]. Such changes have been observed in intervertebral disks [18] as well as in articular cartilage [13]. Though several studies have demonstrated the diversity of physicochemical properties and functions of FMOD, the molecular basis of this structural and functional diversity of FMOD remains unexplored. The role of proteoglycans and more precisely of glycans and GAGs moieties is still difficult to decipher [19]. Until recently, no direct biophysical characterizations were used. However, in a previous study, Angarra et al [20] were able to access to a direct observation of glycans using a combination of ions electrospray, low-temperature scanning tunneling microscopy and quantum DFT calculations. However, this new approach remains too recent to treat all categories of molecules. In this context, numerical simulations using molecular dynamics is a very informative and useful method [21–23]. Furthermore, the dynamics (e.g., conformation evolution over time) of proteoglycans such as FMOD and the role of its carbohydrate residues have not yet been investigated and different scenarii of molecular systems may be investigated. In this study, an attempt was made to demonstrate how carbohydrate chains influence proteoglycan dynamics. Molecular dynamics simulations of FMOD in three configurations were performed: non-glycosylated FMOD, glycoprotein with complex type glycans and proteoglycan with keratan sulfate GAGs. How variation in carbohydrate composition influences various FMOD properties was assessed; protein dynamics, conformational preferences, solvent accessibilities of important residues, to name a few, were evaluated.

2 | Materials and Methods

2.1 | Modeling FMOD Variants

The FMOD X-ray crystallographic structure (PDB: 5MX0) was used as the starting structure for further modeling studies [16]. Since the experimental structure lacks the N-terminal domain, the missing N-terminal residues were modeled using CHARMM-GUI PDB reader server [24–26]. Residues constituting the signal peptide were ignored, and residues located after the signal peptide were modeled. Thus, FMOD residue 15 was renumbered as residue 1 in this work, and subsequent residues were renumbered accordingly. All FMOD residues in this work will be referenced according to the renumbered scheme, unless stated otherwise.

Tyrosine sulfate residues in the N-terminal domain (residues 20, 38, 39, 45, 47, 53, and 55; residues 6, 24, 25, 31, 33, 39, and 41 in this work) were modeled from the sulfotyrosine structure in the PubChem database [27] (PubChem compound ID: 514186). In this work, the sulfated tyrosine residues are named Tys. These residues were inserted in the FMOD structure using the UCSF Chimera program [28]. Carbohydrate moieties were added to residues 113, 152, 187, and 277, in accordance with the carbohydrate attachment sites characterized from the X-ray crystallographic structure [16]. For the proteoglycan, the keratan sulfate-1 type GAGs were modeled (33 saccharide residues/chain), and complex-type glycans (12 saccharide reisdues/chain) were modeled using the Glycam web server [29]. The glycoprotein form of FMOD was modeled with complex-type core-fucosylated glycans including sialic acid cap. The composition of Keratan sulfate GAGs were adopted from mass-spectroscopy studies performed on FMOD proteoglycan from different tissues [30, 31]. Figure 1 provides all the FMOD systems simulated in this study.

2.2 | MD Simulations

All the following simulation preparation and production steps were performed using the AMBER18 program [34, 35]. The parameters described below were applied to all the three FMOD systems, unless otherwise stated. Based on the modeled FMOD structures, protein residues were described using the AMBER ffsb14 force field [36]. Carbohydrate residues were described using Glycam_06j-1 force field [37]. Energy parameters

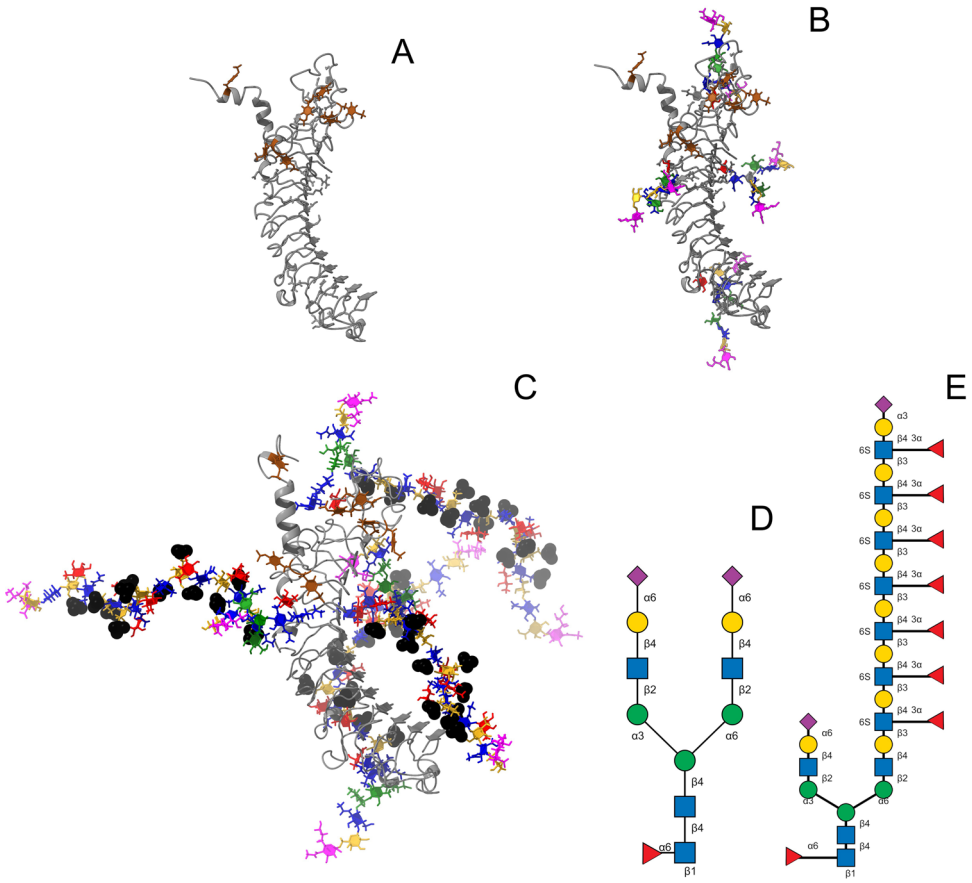


FIGURE 1 | Structures of the FMOD systems studied in this work. Representation of FMOD in non-glycosylated (A), glycoprotein (B) and proteoglycan states (C). Protein residues are represented as cartoons, residues of carbohydrate chains are represented as sticks colored according to the SNFG scheme [32]. Sulfate ions are represented as black spheres, the tyrosine sulfate residues are represented as orange sticks. (D & E) Schematic representation of the glycan (D) and GAG chains (E) modeled on FMOD. The carbohydrate residues are colored according to the SNFG scheme [32]. The structure and schematic representations were generated using UCSF Chimera and GlycoGlyph tools respectively [28, 33].

describing the tyrosine sulfate residues were derived using the Antechamber program of the AMBER molecular dynamics program [38]. The charges to the tyrosine sulfate molecule were assigned based on the AM1-BCC model [39]. The system was solvated in an octahedral box, with an edge length of 20 Å. The water molecules were described using the TIP3P model [40], and neutralizing sodium chloride ions of 0.1 M concentration were added to the systems. The simulation boxes contained 198,438, 242,156, and 341,834 atoms in the case of non-glycosylated FMOD, glycoprotein and proteoglycan, respectively. The solvated systems were subjected to energy minimizations for 100,000 steps using the steepest descent algorithm. After minimization, equilibration in the NVT ensemble was performed for 1 ns to bring the temperature to 100 K. The integration time step was set to 0.002 ps. A second NVT equilibration was run for 1 ns, with a time step of 0.002 ps to reach the temperature of 310 K. A final equilibration in NPT ensemble was performed for 2 ns, still at a temperature of 310 K. Temperature and pressure were maintained using the Berendsen thermostat and barostat, respectively [41]. The bonds involving hydrogen atoms were constrained using SHAKE algorithm [42]. During the equilibration steps, positional restraints were imposed on the heavy protein and carbohydrate atoms. After equilibration, production simulations were carried out in four replicates, each replicate consisting of a 500 ns sampling. In each replica, the time-step was set to 0.002 ps, temperature and pressure were maintained at 310 K and 1 atm respectively. The cutoff threshold for nonbonded interactions was set at 13 Å, and long range interactions were calculated using Ewald method. Simulations on the three FMOD systems resulted in a grand total sampling of 6μs. Simulation details are summarized in Supporting Information S1: Table 1.

2.3 | Processing of Trajectories for Analyses

Once the simulations had been completed, the trajectories were processed in such a way as to center the FMOD molecule within the box for each snapshot. The four replicas are concatenated into a sub-trajectory that retains one snapshot in 100 (corresponding to a time interval of 1 ns). All analyses were performed on these refined trajectories.

2.4 | FMOD Protein Dynamics Analyses

To assess the overall changes in protein as well as carbohydrate conformational preferences, The RMSD of FMOD c-alpha atoms were calculated from each replicate trajectories and compared; similar calculations were also performed on the

GAG chains as well as glycan chains from all the replicate trajectories. To evaluate the flexibility of FMOD residues for each of the structure, Root Mean Square Fluctuations (RMSF) were calculated for each trajectory, considering the C-alpha atoms of each residue. To evaluate the essential global dynamics associated with each FMOD state, a Principal Component Analysis (PCA) was performed. Two types of PCA calculations were performed: in the first type, PCA was performed on the C-alpha atoms of the LRR domains (residues 74–362, corresponding to the LRR N-Terminal, LRR1-11 and LRR C Terminal domains) of each FMOD trajectory, corresponding to each glycosylation state. From this PCA calculation, the dynamics associated with the first eigenvector were extracted and visualized. PCA calculations were performed using the ProDy program [43], and visualization was performed using the Normal Mode Wizard of ProDy program and the VMD program [43, 44]. In the second type of PCA calculation, an ensemble of FMOD trajectory conformations was created by concatenating all the three FMOD trajectories. On this set, a PCA was performed on the C-alpha atoms of the full-length FMOD, using Bio3D MD analysis program [45, 46]. FMOD conformations were projected onto the subspace of first and second principal components, the two together accounting for 52.7% of the trajectory ensemble. From this projection, a hierarchical k-means clustering analysis was performed, and the FMOD conformations were grouped into different clusters. From each cluster, the average conformation was extracted, and conformations from different clusters were superimposed and visualized using the UCSF Chimera program [28]. To evaluate the relative Gibbs free energy associated with each cluster, the coordinates of the projections on the first and second principal components were extracted, and the relative Gibbs free energies were calculated from these coordinates using the gmx sham tool of the GROMACS program [47, 48].

2.5 | Analyses of Relative Solvent Accessibility of FMOD Residues

To assess the solvent accessibility of FMOD residues in the presence of glycans and GAGs, the solvent accessibility of residues involved in FMOD function, such as collagen binding (LRR7, residues 233–242; LRR11, residues 332–343), MMP13 cleavage site (residues 49–50) and the N-terminal helical region (residues 24–26) was calculated. The absolute solvent-accessible surface areas were averaged for each residue over the entire trajectory for glycoprotein and proteoglycan states. Calculations were performed using the gmx sasa program of GROMACS [47, 48]. For all the solvent accessibility calculations mentioned below, the probe radius was set to 0.14 nm (1.4 Å). From the average solvent-accessible surface areas, the relative solvent accessibilities for each residue were calculated according to Equation (1):

$$RSA_i = \frac{ASA_i}{ASA_{i'}} \cdot 100 \quad (1)$$

Where RSA_i denotes the relative solvent accessibility of residue i, ASA_i denotes the absolute solvent-accessible surface area of residue i calculated from the simulation trajectories, $ASA_{i'}$ denotes the absolute solvent-accessible surface area of residue i in a "GiG" tripeptide consisting of residue i flanked by glycine residues on either side. The $ASA_{i'}$ values were taken from the solvent accessibility studies carried out by Miller et al. [49] To calculate the accessibility of tyrosine sulfate residues, a tripeptide consisting of tyrosine sulfate flanked by glycine residues on the either side was modeled using the UCSF Chimera program and the CHARMM-GUI PDB reader program [24–26, 28]. The absolute solvent-accessible surface area was calculated using the gmx sasa program of GROMACS [47, 48].

2.6 | FMOD Residue Conservation Analyses

To assess the conservation of FMOD residues, a multiple sequence alignment of FMOD orthologs was performed. All FMOD homologs were searched using the phmmer web server [50], using the human FMOD sequence (Uniprot ID: Q06828) as query. The search yielded > 5000 homologous sequences, from which 621 sequences annotated as FMOD were filtered. The 621 FMOD sequences were aligned with the MAFFT multiple sequence alignment program, using the L-INS-i method [51, 52]. Aligned sequences were visualized and residue conservation was calculated using the Jalview program [53].

3 | Results

3.1 | Changes in Carbohydrate Composition Impact FMOD Dynamics

To assess changes in protein and carbohydrate dynamics associated with each state, we calculated C-alpha as well as carbohydrate RMSD of each replicate and plotted them. Similarly, the snapshots of every 50 ns of from each replicate was taken, and the snapshots were superimposed and compared for each replicate. We found that most of the replicates have similar RMSD range, suggesting that the sampled conformational space and dynamics could also be similar (Supporting Information S1: Figure 1). This was also reflected in the snapshots from both proteoglycan as well as glycoprotein simulations, where the FMOD conformations have a uniform pattern in both the proteoglycan as well as glycoprotein states, in addition to the glycan and the GAG conformations in respective states. The GAG conformation of the replicate-4 is a notable exception, where the GAG conformational preference is substantially different from the one observed in other three replicates (Supporting Information S1: Figure 2). This is also accompanied with the interactions with protein residues of the N-terminal helix region. Consequence of this is also reflected in high C-alpha RMSD of FMOD proteoglycan replicate-4, which could be due to potential destabilization of the N-terminal helix as a result of interaction with charged GAGs.

To assess FMOD dynamics of each residue in their respective states, C-alpha RMSF of FMOD residues were calculated from the respective simulation trajectories. To determine if a given residue is rigid or flexible, a threshold RMSF value of 0.3 nm (3 Å) was considered. Two patterns of FMOD dynamics were observed: the region consisting of LRR domains was found to be

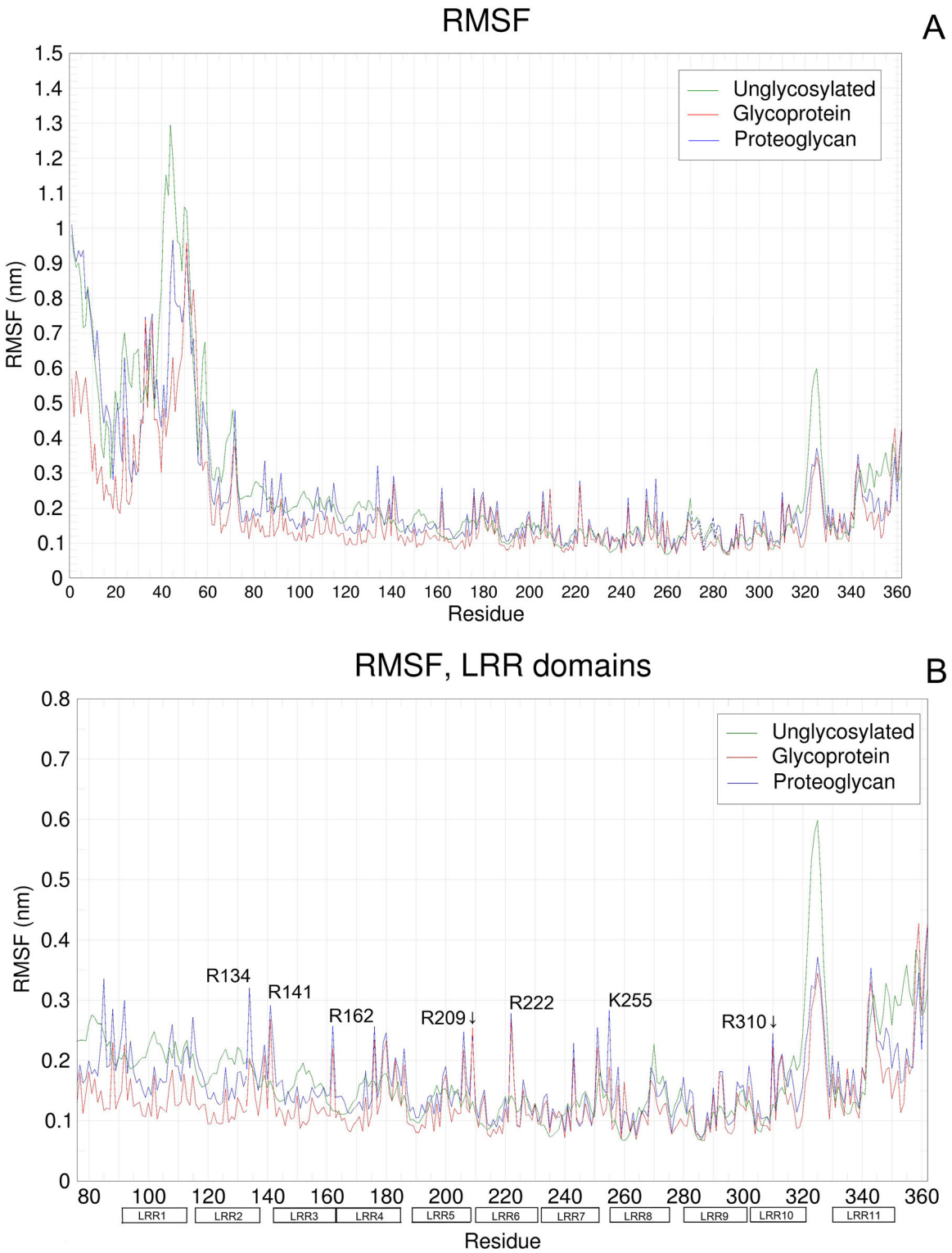


FIGURE 2 | Evolution of Root Mean Square Fluctuations (RMSF) of C-alpha atoms in Fibromodulin. Panel A shows the evolution of the RMSF along the complete sequence, while panel B shows this evolution along the LRR regions. The RMSFs of FMOD in the non-glycosylated, glycoprotein and proteoglycan states are represented by green, red and blue lines, respectively. RMSFs were calculated using the gmx rmsf tool of GROMACS [47, 48].

mostly rigid. The average C-alpha RMSF was between 0.1 and 0.3 nm (1–3 Å). On the other hand, N-terminal domain region was highly flexible, with an average RMSF of up to 1.3 nm (13 Å) (Figure 2A). The LRR domains are largely rigid in all three FMOD states, with an RMSF ranging between 1 and 2 Å (Figure 2B). Interestingly, the N-terminal region in the proteoglycan state is more flexible than in the glycoprotein state, with RMSFs of some residues higher by margin of about 1 Å on average, and up to 4 Å for some residues. The non-glycosylated state, on the other hand, shows high flexibility for

all residues in the N-terminal region. A similar pattern is also present for the N-terminal LRR domains, spanning from residues 80–140 (Figure 2A, B). Interestingly, comparison of RMSF in proteoglycan and glycoprotein states represented as B-factors, with the B-factors of the X-ray crystallographic structure suggests some differences in the patterns of FMOD dynamics. Though the overall dynamic patterns are conserved across the MD simulations and the crystallographic structure, the residues around the second glycosylation position in FMOD crystallographic structure is more flexible as compared to the residue dynamics in the MD simulations (Supporting Information S1: Figure 3). Similarly, the N-terminal LRR domain in both proteoglycan as well as glycoprotein states are more flexible, than in the X-ray crystallographic structure (Supporting Information S1: Figure 3). Though the differing dynamics patterns in experiments versus MD is worth further investigation, it is difficult to comment at this point if the differences in dynamics could be attributed to the sampling gaps in the MD simulation or if they could be crystal-packing artefacts. Furthermore, the influence of carbohydrates towards this is also unclear, as it is unclear if the FMOD in the crystallographic studies is in the proteoglycan or in the glycoprotein configuration.

Surprisingly, though LRR domains are rigid globally, some residues in LRR regions are more flexible in glycosylated states (proteoglycan & glycoprotein states) than in non-glycosylated state. In particular, two observations can be made: (1.) all these residues are flexible in the glycoprotein and proteoglycan states, while they are relatively rigid in the non-glycosylated state. (2.) all flexible residues, without exception, are positively charged residues, and are located at the convex face of the curved solenoid (Figure 2B). Though the leucine rich repeat domains are generally considered to have a rigid and conserved curved-solenoid structural architecture [54], such local flexibility of residues observed here suggests their potential roles in modulating protein-protein interactions [55, 56].

The large difference between FMOD dynamics in N-terminal region and the LRR domains is consistent with observations from experimental studies as well as studies of FMOD structure. But such variation in FMOD dynamics poses a challenge for understanding its conformational preferences. Thus, to evaluate the essential dynamics of different forms of FMOD, two types of Principal Component Analysis (PCA) were carried out. In the first type, PCA was performed on the LRR domains (LRR1-C-terminal) of FMOD, to understand how LRR domain dynamics varied in MD trajectories, whereas in the second type, essential dynamics were evaluated for the full-length FMOD, including LRR domains and the N-terminal disordered region. PCA on LRR domains revealed substantial variation in FMOD dynamics between the non-glycosylated, glycoprotein and proteoglycan states. In the non-glycosylated state, the LRR domains remained largely rigid; indeed, in PC1 of this state, none of the LRR residues forming the curved solenoid had fluctuations greater than 2 Å (Figure 3). In contrast, in the glycoprotein state, the N- and C-terminals of the curved solenoid domain underwent twisting dynamics. Finally, in the proteoglycan state, the curved solenoid domain of the FMOD undergoes stretching dynamics, resulting in deformation of the curved solenoid architecture (Figure 3).

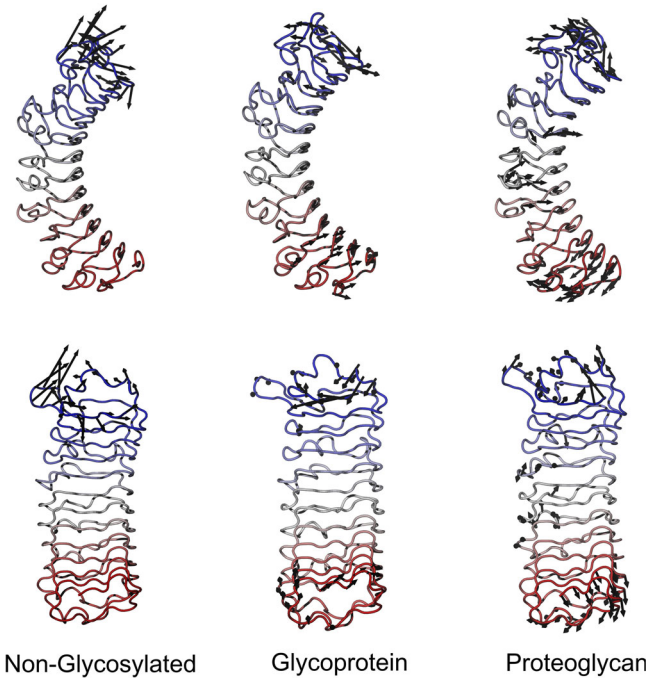


FIGURE 3 | Essential dynamics of Fibromodulin in non-glycosylated, glycoprotein and proteoglycan states observed from the side (top) and from the front (bottom). The directions of residue dynamics whose fluctuations are greater than 2 Å are represented by an arrow. The dynamics associated with PC1 account for 24.1%, 15.6% and 20.6% of the trajectory in the non-glycosylated, glycoprotein and proteoglycan states respectively. Essential dynamics calculations were performed using the ProDy program [43] and visualized using the VMD program [44].

3.2 | A Rigid, Conserved Region in the N-Terminal Region Was Identified as Potentially Modulating FMOD Functions

In the second set of PCA calculations, analyses were performed on an ensemble of all the full-length FMOD conformations. The conformations were projected onto the conformational space of principal components 1 and 2, representing around 52% of all FMOD conformational sampling. The three FMOD states were grouped into different clusters which were unique for each state (Figure 4A). There was no overlap between the conformations of proteoglycan and glycoprotein, indicating that the conformational preferences of both the states are unique and mutually exclusive. (Figure 4A). Though some overlap was observed between glycoprotein and non-glycosylated conformations, non-glycosylated conformations exploring the subspace of glycosylated conformations do not form a cluster. Instead, another cluster is formed that is distinct from the glycoprotein conformation, which does not exhibit the energy minima observed in the glycoprotein cluster. This suggests that glycoprotein carbohydrates play a significant role in stabilizing glycoprotein conformations. As a result of this stabilizing effect, a distinct cluster corresponding to the glycoprotein conformations is formed and exhibits the lowest energy minima among the different clusters (Figure 4B).

The free-energy landscape of the FMOD conformations was plotted using the projections of the FMOD conformations with

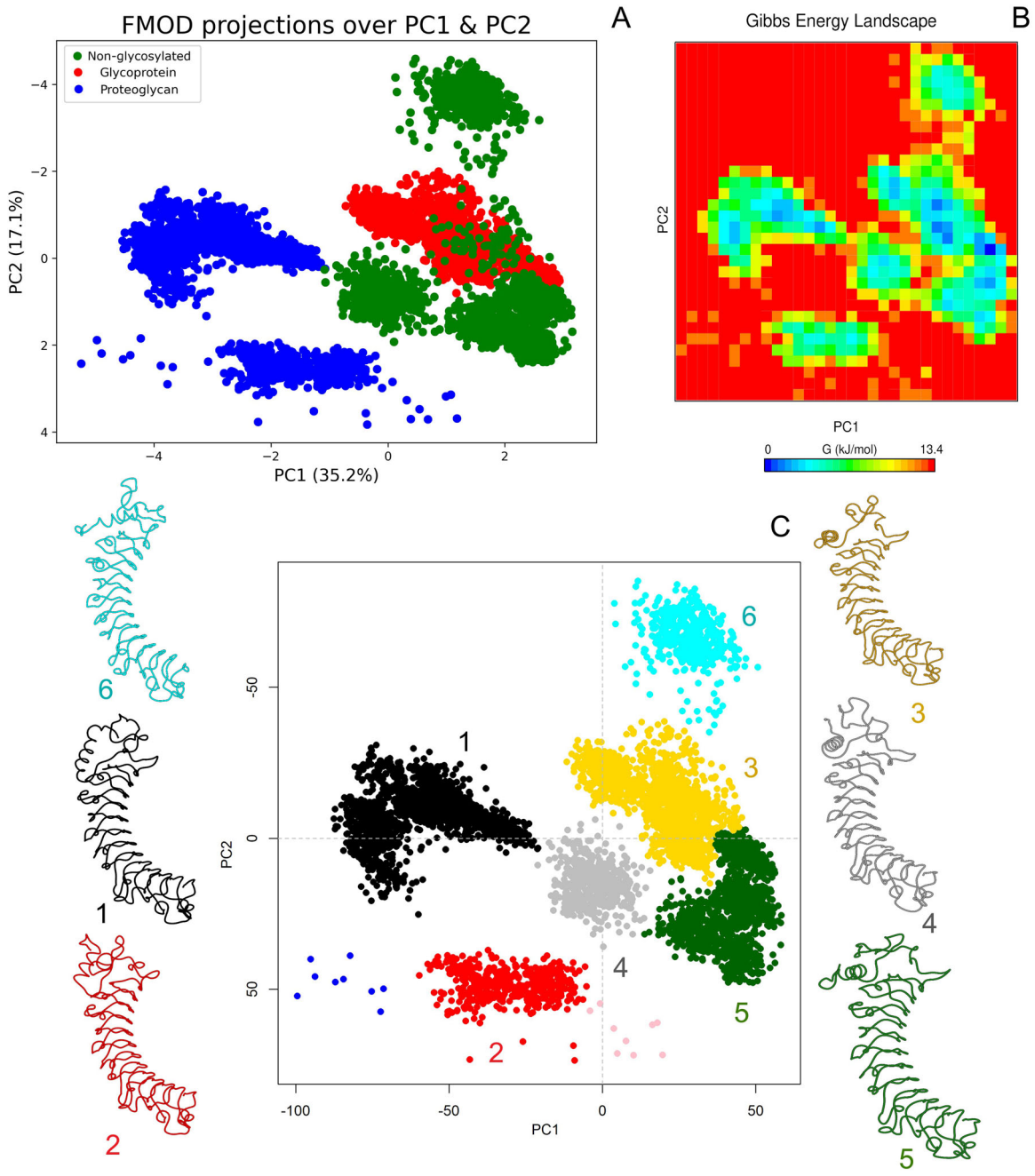


FIGURE 4 | PCA applied on the different states of FMOD. (A) Projections of FMOD conformations onto the subspace of principal components 1 and 2. The FMOD conformations of each state are represented in different colors. Principal components 1 and 2 account for 32.5% and 20.1% of all the FMOD conformations respectively. PCA calculations were performed using the ProDy program [43]. (B) Projections of conformations in (A) described as a free-energy landscape. Relative Gibbs free energy is represented in terms of a color gradient from blue (low energy) to red (high energy). Free energy landscape calculations were performed using the gmx sham program of GROMACS. (C) Projections in (A) represented as clusters representing different FMOD conformations. Clustering was performed using the k-means method, and the average conformation associated with each cluster was extracted, superimposed and visualized. PCA calculations, clustering and average conformation extractions were performed using the Bio3D program [45, 46], and representative conformations were visualized using the UCSF Chimera program [28].

respect to the first and second principal components as coordinates. This revealed the presence of multiple energy minima associated with different conformations (Figure 4B). These minima are distributed between proteoglycan and glycoprotein conformations. No energy minima are observed in non-glycosylated conformations. The conformations of the glycoprotein and proteoglycan states exhibit distinct energy wells; the deepest well is associated with the glycoprotein conformational cluster, suggesting that the lower-energy glycoprotein conformations exhibit greater stability than the proteoglycan conformations (Figure 4B).

Another objective of the PCA was to examine the structures representing conformational clusters. To this end, a hierarchical k-means clustering calculation was performed on the FMOD projections on PC1 and PC2, which revealed 6 different clusters

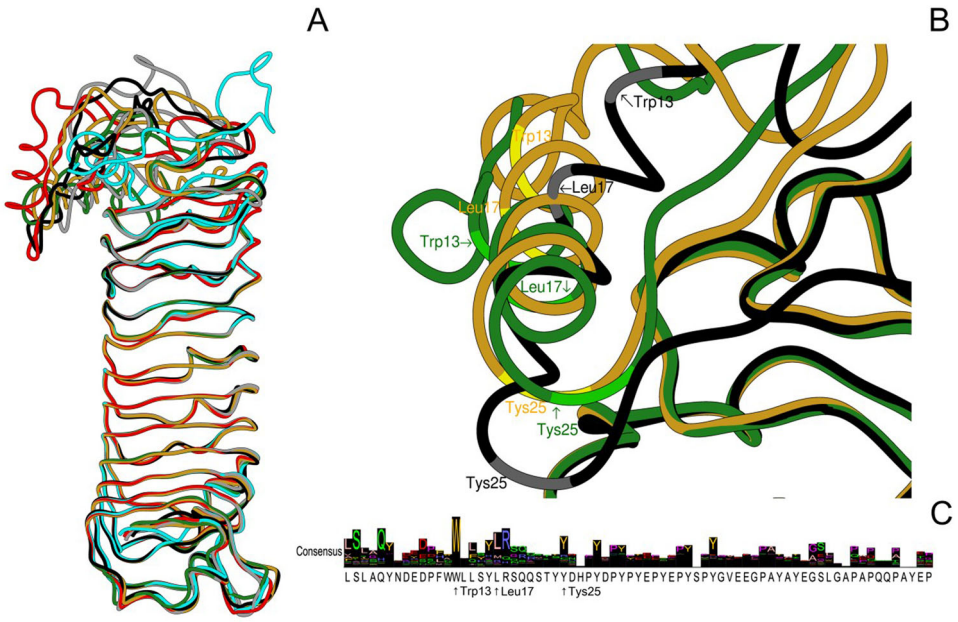


FIGURE 5 | (A) Superimposition of the representative average FMOD conformations shown in Figure 4C, obtained from k-means clustering calculations of FMOD trajectory ensemble. (B) Superimposition of the representative conformations of clusters 1, 3, and 5 from Figure 4C, corresponding to proteoglycan, glycoprotein and non-glycosylated conformations. (C) Results of multiple sequence alignment performed on the N-terminal region of the 621 FMOD orthologs. Tys corresponds to the modified tyrosine sulfate residue.

across the three FMOD states (Figure 4C). Distance-based clustering calculations were performed because it is difficult to characterize full-length FMOD dynamics in the same way as for LRR domains (Figure 4C), due to the high flexibility associated with the disordered N-terminal domain (Figure 2A). Non-glycosylated FMOD corresponded to three unique clusters, while FMOD in glycoprotein form corresponded to one unique cluster and FMOD in proteoglycan form corresponded to two unique clusters (Figure 4C). From the obtained clusters, the average conformations corresponding to each of the clusters were extracted and the representative conformations were structurally superimposed. Extracting the average representative structures and comparing them shows that, while the LRR domains are rigid and do not exhibit large conformational variations, most conformational variations are localized to the N-terminal disordered region (Figure 5A).

A structural feature common to all clusters is the presence of a kink/bend between residues 24–26 (Tys-Tys-Asp; Figure 5A; Tys corresponds to the modified tyrosine sulfate residue). This motif is composed solely of negatively charged residues. Interestingly, RMSF calculations show that the residues immediately preceding this motif are relatively rigid compared to the other residues in the N-terminal disordered region, with an RMSF between 2 and 4 Å (Figure 2A). Though the N-terminal disordered domain sampled a variety of conformations (Figure 5A), the presence of the helix followed by the kink is conserved in all conformations (Figure 5B). This suggests that this structural arrangement and the residues in this region could be crucial for protein-protein interactions of FMOD with many other partners. To evaluate this hypothesis, a sequence search for different FMOD orthologs and a multiple sequence alignment of FMOD orthologs were carried out, with the aim of assessing the conservation of residues in this motif. The alignment of 621 FMOD orthologs showed that the residue tyrosine sulfate forming the bend (Tys25) is well conserved in 62% of 621 FMOD orthologs (Figure 5C; Supporting Information S1: Table 2). In addition, the residues preceding this amino acid form a helix and are also highly conserved (Figure 5C; Supporting Information S1: Table 2), and the protein complex interface prediction by hybridPBRpred web-server [57] show that the residue Tyr24 is predicted to have high propensity for forming the protein complex interface (Supporting Information S1: Table 3). These observations support the hypothesis that due to formation of a unique structural scaffold that may aid in "docking" the interacting partner, this motif may play an important role in FMOD's interactions with various protein partners, along with the disordered regions in its vicinity.

3.3 | Significant FMOD Residues Remain Accessible in Proteoglycan/Glycoprotein States

The proteoglycan form of FMOD has a longer GAG chain than the glycan chain of its glycoprotein form. Indeed, visualization of the carbohydrate conformational distribution indicates that the carbohydrates have sampled a large conformational space, potentially forming contacts with numerous FMOD residues. This was particularly true for the FMOD proteoglycan, which shows that very few FMOD residues were not in close proximity to the carbohydrates during the simulations (Figure 6). It was therefore hypothesized that there might be a negative impact on the solvent accessibility of FMOD residues in the proteoglycan state. To test this hypothesis, the solvent-accessible surface areas, as well as the absolute and relative solvent accessibilities of important FMOD residues in the proteoglycan MD simulations, were evaluated. Calculations were performed on four sets of residues: the LRR7 and LRR11 residues involved in collagen

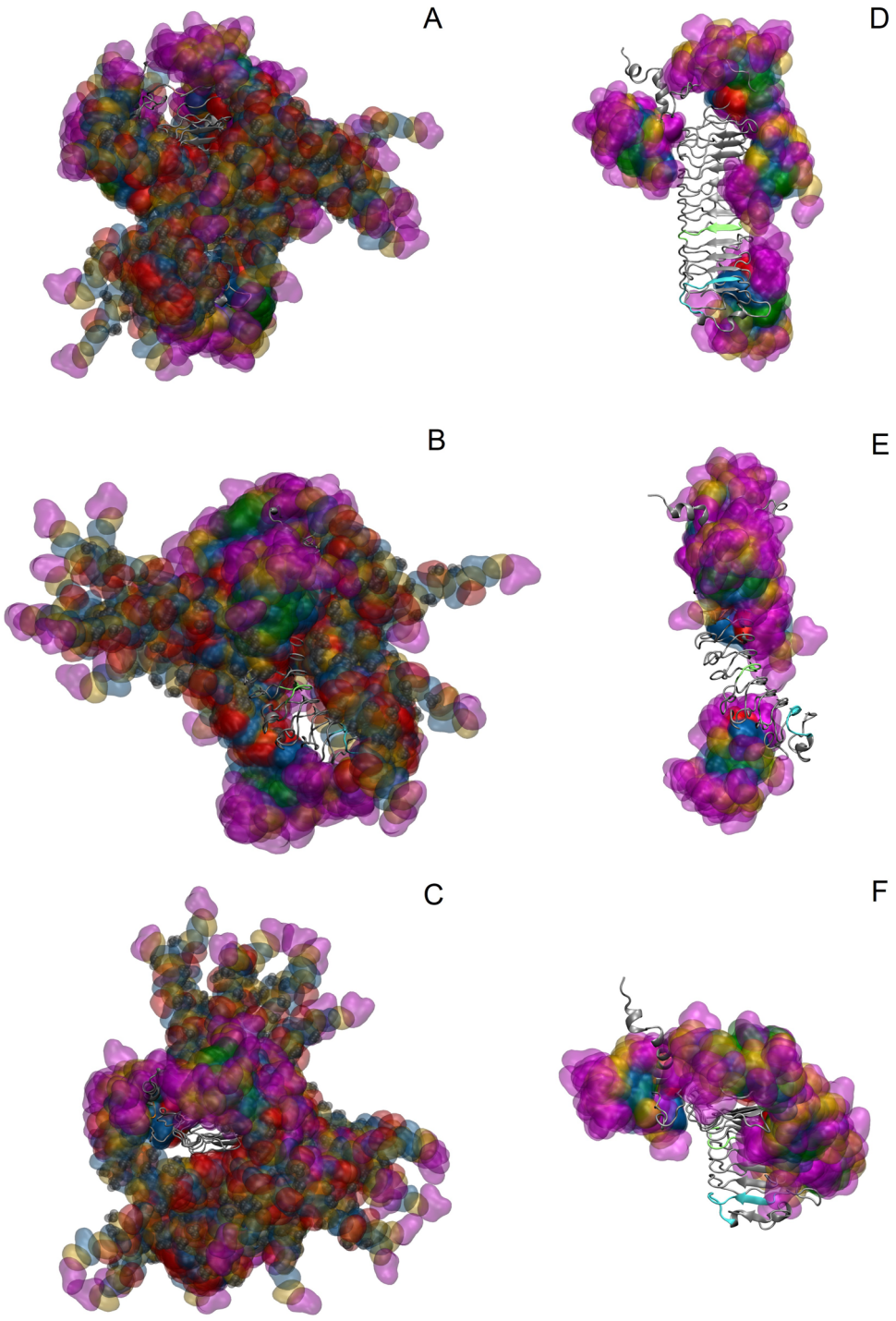


FIGURE 6 | Front, side and top view of FMOD proteoglycan (A, B, and C respectively) and glycoprotein (D, E, and F respectively) showing the distribution of carbohydrate conformations in the simulation. Carbohydrate conformations were taken every 100 ns and projected onto the FMOD structure. The protein is represented as silver ribbon, carbohydrate residues are represented as surfaces, and colored according to the SNFG convention [32]. Sulfate groups are represented by gray spheres. The collagen-binding regions LRR7 and LRR11, and the MMP-13 binding residues are represented as light green, cyan and light pink ribbons respectively. Conformations were visualized using the VMD program [44].

binding [58, 59], the MMP13 cleavage site located on residues 49 and 50 at the N-terminal domain [15], and residues 24–26 located at the end of the N-terminal helix, which, in the simulation, are observed to form a "hinge" in the N-terminal dynamics. Absolute accessible surface areas were averaged for all residues over all the frames of the trajectory, and relative accessible surface areas were derived from the calculated average absolute surface areas.

In LRR7, differences in relative accessibility were found to be rather small in both the proteoglycan and glycoprotein states. However, with the exception of certain hydrophobic residues (Leu235, Met237, Val242), most of residues remain accessible with a relative accessible area > 20. Most residues, with the exception of Gly233, exhibited a slight decrease in relative accessibility in the proteoglycan state compared to the glycoprotein state, though it did not alter the overall state of

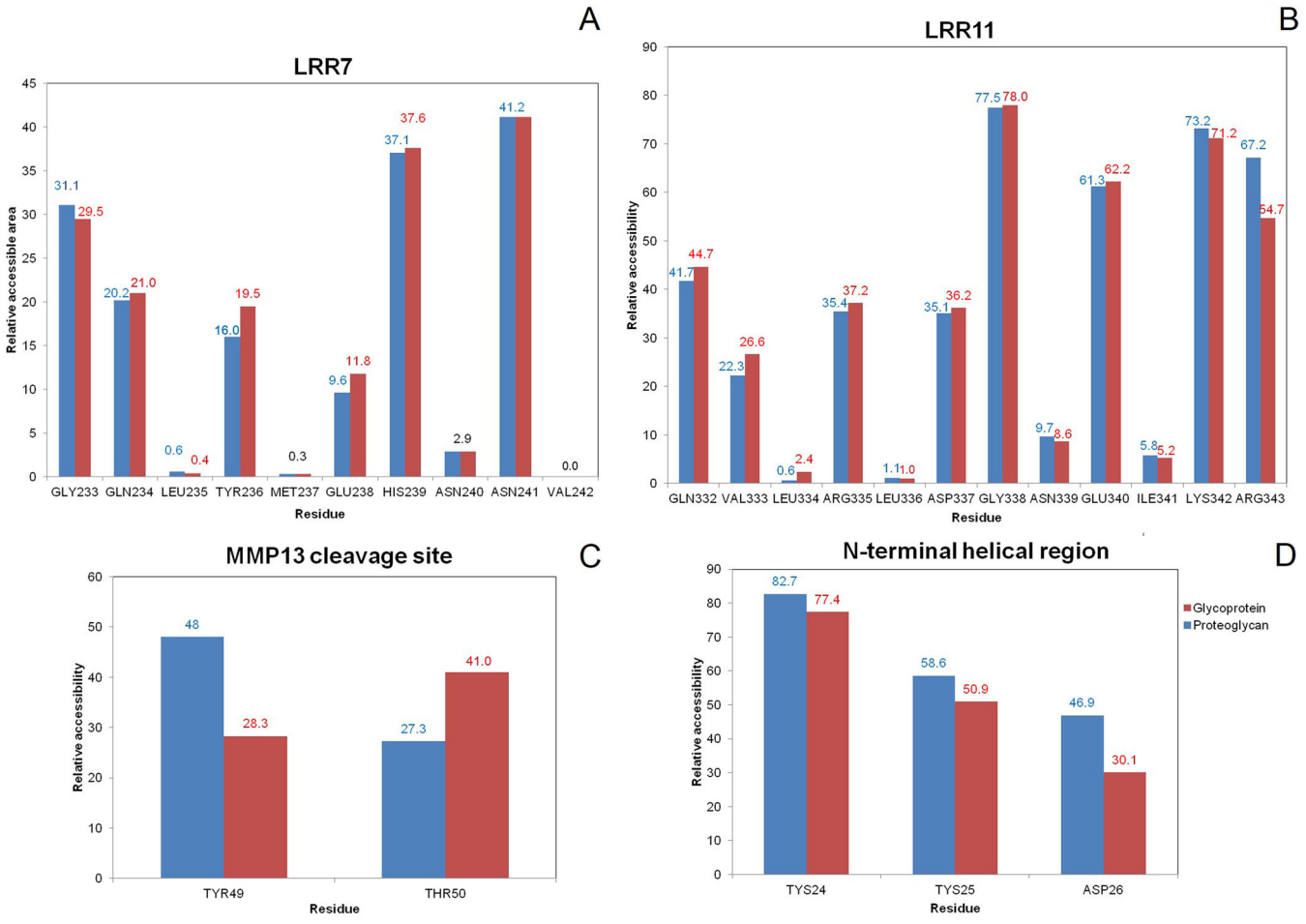


FIGURE 7 | Plot showing the relative solvent-accessible surface areas of residues constituting LRR7 (A), LRR11 (B), the MMP13 cleavage site (C) and the loop following N-terminal helical region (D). Absolute solvent-accessible surface areas were calculated as the average of the areas of all conformations in the trajectory, using the gmx sasa program of GROMACS [47, 60]. Relative accessible areas were calculated as described in the methods. Tys corresponds to the modified tyrosine sulfate residue.

accessibility, that is, residues continued to remain accessible even if absolute accessible areas decreased slightly (Figure 7A). However, the accessibility of some residues proved to be low. The residue Glu238 was found to be buried in both proteoglycan as well as glycoprotein states, with relative solvent accessibilities of 9.6 and 11.8 in proteoglycan and glycoprotein states respectively (Figure 7A).

Similarly, many LR11 residues were also accessible. In particular, residues Glu340 and Lys342 are observed to have high relative accessibility in both proteoglycan and glycoprotein states (Figure 7B). These two residues were found to be critical for FMOD binding to collagen triple helices in previous experimental studies [58].

In addition to the collagen-binding sites, relative accessibility calculations were performed at the MMP-13 cleavage site, located at residues Tyr49 and Thr50 in the N-terminal domain. These residues were found to be the site where MMP-13 bound to the FMOD and proteolyzed, resulting in the loss of collagen fibril organization in osteoarthritis [15]. Based on accessibility calculations, mixed results were observed: though the relative accessibility of residue Tyr49 decreased significantly in the glycoprotein versus proteoglycan state, the relative accessibility of Thr50 increased similarly in the glycoprotein versus proteoglycan state (Figure 7C). Though both residues showed contrasting variations in relative accessibility in the two simulations, they remained accessible in both proteoglycans and glycoproteins, with relative accessibility > 20. The absolute accessible area of the cleavage site comprising the two residues was also calculated. The absolute accessible area decreased slightly in proteoglycans compared with glycoproteins (Figure 7C). Due to this small difference in accessible area at the cleavage site, it is not possible to conclude whether or not carbohydrate chains have significant impact on the accessibility of the MMP-13 cleavage site. Given the value of relative accessibility for these residues, it seems relevant to consider that FMOD remains sensitive to MMP-13 cleavage in both states.

It was observed that, although the N-terminal domain is highly flexible, certain conformations are common to all clusters in all states (Figure 5A, B). This conformation involves formation of a turn immediately after the helix in this region, followed by a disordered coil region. The bend appears to act as a hinge, influencing the dynamics of the surrounding regions. It was further hypothesized that the region might also be involved in several protein-protein interactions due to its rigid nature, which may be accompanied by a high relative solvent

accessibility of the residues comprising it. Solvent accessibility calculations showed that the three residues forming the turn were highly accessible in both the proteoglycan and glycoprotein states. In the proteoglycan state, the three residues consisting of tyrosine sulfate and aspartate had relative solvent accessibility of 62, 58, and 47 respectively. However, there is an overall global decrease in the relative accessibility of these residues in the glycoprotein state (Figure 7D). Though the residues do not lose their accessibility entirely, reduced accessibility suggests greater obstacles to protein-protein interactions involving these residues.

4 | Discussion

Though FMOD is a versatile protein interacting with a wide variety of proteins such as collagen, LOX, MMP-13, integrins [8, 12, 15, 58] and its involvement in various processes including development, immune response and cancers, how differences in its carbohydrate composition influences its conformations and ultimately functions remained a mystery. To improve this understanding, molecular dynamics simulations of FMOD in different glycosylation states have been carried out. RMSF calculations reveal a global trend whereby FMOD in proteoglycan form is marginally more flexible than in glycoprotein form (Figure 2). A possible cause for this phenomenon could be electrostatic repulsion resulting from interactions between negatively charged FMOD residues and sulfate groups carried by the GAG chains. On the other hand, in the presence of interaction partners comprising positively charged motifs, FMOD's interactions with such partners may further stabilize the proteoglycan form by increasing the contribution of attractive contacts to the overall free energy. This hypothesis could be tested in future studies by docking FMOD to partners such as collagens and integrins. Furthermore, in proteoglycan, some positively charged residues within the LRR domains are also flexible. Although such flexibility is surprising, its significance is unknown. It is also interesting to note that, based on X-ray crystallography studies [16], many positively charged residues, particularly histidines, were observed to be solvent-accessible within the structure. It was hypothesized that these residues might be involved in protein-protein interactions. It is also possible that the flexibility of these residues facilitates protein-protein interactions.

It was also observed that the N-terminal region of the non-glycosylated FMOD is destabilized, with a higher RMSF compared to both states of the glycosylated FMOD (Figure 2A). Similar observations were also made from the evaluation of the global dynamics through the PCA calculations and the free-energy landscape calculations derived from the coordinates of the first and second eigenvectors of the PCA calculations (Figure 4B). Though different energy wells were observed in the proteoglycan and glycoprotein conformations of FMOD, the non-glycosylated FMOD conformations were found to be highly dispersed, accompanied by an absence of free-energy minima. It is therefore clear that carbohydrates have a significant stabilizing effect on FMOD dynamics. Furthermore, this observation is consistent with another previous study conducted on various glycosylated proteins from the Protein Data Bank, which reached the same conclusion [61]. Relative free energy calculations also revealed that proteoglycan and glycoprotein forms exhibit multiple energy wells associated with their conformations. The deepest energy minimum was associated with a cluster belonging to glycoprotein conformations (Figure 4B). This is also consistent with observations from RMSF calculations (Figure 2A) and may be due to repulsive electrostatic interactions between negatively charged FMOD residues and GAG sulfate groups. Characterization of the dynamics of the first principal component of the three states revealed that the LRR domains of FMOD have different overall dynamics (Figure 3). These dynamics could be involved in potential FMOD interactions with other protein partners by regulating the accessibility of LRR residues. This also shows that, though LRR domains appear to be rigid, there are substantial dynamics involving these domains. This can also be observed from the RMSF of LRR domain residues in proteoglycan and glycoprotein states (Figure 2C).

It was also found that the N-terminal domain is disordered and highly dynamic in all FMOD states. Despite this, there were dynamics patterns common to all FMOD states. These included the formation of a hinge immediately following a helix motif. Interestingly, this region is entirely composed of negatively charged residues, namely tyrosine sulfate and aspartate residues. Studies have also suggested that the negatively charged residues in the N-terminal domain, including tyrosine sulfate residues, have the potential to bind a wide variety of other extracellular matrix proteins that have positively charged surface patches, such as fibroblast growth factor-2, thrombospondin-I, interleukin-10, and so on, as well as numerous other proteins such as LOX, TGF-β, IL1-receptor type-I and so on [17]. Since the hinge region is rigid compared to the rest of the N-terminal region and is solvent-accessible in the proteoglycan and glycoprotein states (Figure 7D), this region could be involved in the protein interactions discussed by Tillgren et al. It was further observed that, from a sequence point of view, the hinge motif, as well as the N-terminal helix residues Trp13, Leu17, Arg18 and Tys25, are well-conserved on a large number of FMOD proteins. Moreover, these residues are located on the outer face of the helix in the proteoglycan and glycoprotein conformations (Figure 5C), leaving them exposed and potentially accessible to other partners.

Since the proteoglycan form of FMOD consists of GAGs with longer chain length than the glycans of the glycoprotein form, it was hypothesized that the accessibility of several FMOD residues might be affected, which could be a trade-off for the greater water-retention capacity that comes with the GAG chains. Contrary to this hypothesis, many important FMOD residues remained accessible in the proteoglycan state, although the relative solvent accessibility decreased slightly compared to the glycoprotein state (Figure 7A, B). These included the LRR-11 collagen-binding domain and the LRR-7 domain. Residues within LRR-7 were also found to have substantially lower relative solvent accessibility, in agreement with a previous experimental study on the collagen-binding capabilities of FMOD. On the other hand, no significant difference was observed in the accessibility of the MMP-13 cleavage site, suggesting that the cleavage site remains accessible in both the glycoprotein and proteoglycan states (Figure 7C). Interestingly, a previous experimental study noted that upon cleavage by MMP-13, FMOD

lost its collagen fibril-binding activity. This suggests the possibility that another region in the N-terminal region may bind collagens, apart from LRR11 and LRR7 regions. This could also be the hinge region following the N-terminal helix, which was found to be relatively less dynamic in MD simulations (Figure 5). This region also underwent a substantial decrease in relative solvent accessibility in the glycoprotein state, in a trend opposite to that of residues in LRR7 and LRR11 (Figure 7D). Though the residues themselves did not become inaccessible (with relative solvent accessibility < 20), such a decrease in accessibility could potentially negatively impact protein-protein interactions involving this region. One possible reason for this decrease in accessibility could be the absence of charged keratan sulfate GAGs, which would mean that carbohydrate residues are freer to form contacts with the negatively charged hinge region, without the risk of interatomic repulsions.

Interestingly, the Glu238 residue of LRR7 has a low relative solvent accessibility equal to 11.8 and 9.6 within the FMOD proteoglycan and glycoprotein respectively. This observation contrasts with the one of previous experimental study on lumican, which showed that the lumican residue homologous to Glu238 is essential for collagen binding [59] (Supporting Information S1: Figure 4). It was also observed in this study that FMOD LRR7 consisting of Glu238 inhibited lumican binding to collagen. Since another experimental study [58] also showed that FMOD LRR7 binds with a low affinity, it may be inferred that other LRR7 residues may contribute more to collagen binding. Furthermore, the low accessibility of Glu238 observed in this work is in good agreement with the observations by Kalamajski and Oldberg [58], suggesting that the low solvent accessibility of Glu238 could contribute to the weak collagen binding by LRR7.

Similarly, most of LRR11 residues have slightly higher relative accessibility in glycoprotein trajectories. Interestingly, however, residues Lys342 and Arg343 have higher relative accessibility in the proteoglycan state than in the glycoprotein state (Figure 7B), which is consistent with previous studies suggesting that the LRR11 region binds collagen better than LRR7 [58, 59].

5 | Conclusions

The structural biology of proteoglycans is one of the major gaps in extracellular matrix biology, as it remains poorly studied to date. The present in silico study demonstrates how variations in the carbohydrate composition of the FMOD proteoglycan influence its dynamics. In particular, it demonstrates that the proteoglycan and glycoprotein states of FMOD exhibit unique and mutually exclusive conformational preferences, which by exposing certain regions can modulate the interactions at play with other protein partners. While its overall conformational dynamics are unique, the conformational preferences of certain regions, such as the loop/hinge that follows the helix within the N-terminal region, are conserved, suggesting that these regions may be critical for FMOD function, including interactions with ECM proteins such as integrins, LOX, collagens to name a few. Ultimately, it was also described how carbohydrates locally influence the accessibility of surface residues. It is impossible to deal with all the combinations of glycosylation sites, the nature of the glycans and GAGs, and sulfatation patterns, which would lead to a huge number of simulations. However, these observations provide a unique insight into the structural dynamics of proteoglycans and provide some clues as to the significant ways in which the composition of carbohydrate chains can influence protein dynamics and functions.

Author Contributions

R.M.R, M.D. and S.B conceived and designed research. R.M.R performed experiments. R.M.R, M.D. and S.B analyzed data and interpreted results of experiments. M.D. and S.B. supervised the project and managed the resources. M.D. performed funding acquisition. R.M.R prepared figures and drafted manuscript. R.M.R, M.D. and S.B edited and revised manuscript. R.M.R, M.D. and S.B approved final version of manuscript.

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Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.